

AN ARITHMETIC INVARIANT $C(N)$ OF BIANCHI 3-ORBIFOLDS FROM A PER-CHARACTER PRETRACE DECOMPOSITION, A UNIVERSAL SELBERG RESIDUE RATIO, AND A CENTER-RANK INEQUALITY

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ABSTRACT. Let $K = \mathbb{Q}(\sqrt{D})$ be an imaginary quadratic field with fundamental discriminant $D < 0$, and let $Y_K := \mathrm{PSL}_2(\mathcal{O}_K) \backslash \mathbb{H}^3$ be the Bianchi 3-orbifold. We assemble, from three independent unconditional ingredients, a purely arithmetic invariant attached to each integer $N \geq 2$:

$$C(N) := \sqrt{2\pi e \cdot \frac{2}{3} \cdot \frac{9}{10}} \left(1 + \frac{1}{N^2}\right) \in \mathbb{R}_{>0}.$$

The three ingredients are: (I) a per-character orthogonal decomposition of $L^2(Y_K)$ along the genus character group $g(K) := \mathrm{Cl}(K)/\mathrm{Cl}(K)^2$ and the corresponding factorisation of the Selberg pretrace identity into absolutely convergent χ -summands; (II) a universal residue ratio $\xi^*(K) \equiv 2/3$ extracted from the identity term of the same pretrace formula, established by a closed-form Γ -function residue computation and verified to 100 digits across eight discriminants $D \in \{-3, -7, -11, -15, -19, -43, -67, -163\}$; (III) a *Center-Rank* inequality $\mathrm{rk}_2(\mathrm{Cl}(Y_K^{\mathrm{CR}}(N))) \geq v_2(N)$ between the 2-rank of the class group of a canonical imaginary quadratic representative $Y_K^{\mathrm{CR}}(N)$ and the 2-adic valuation of N , together with an unconditional Dirichlet companion verified for all 153 fundamental discriminants $D \in [-500, -3]$, which restricts the set of admissible representatives. The $9/10$ prefactor is the Dijkgraaf–Witten genus-0 ratio $N^2/(N^2 + 1)$ evaluated at the $\mathrm{SU}(3)$ anchor. The paper establishes the three ingredients under explicit hypotheses and assembles them into a calculable invariant $C : \mathbb{N}_{\geq 2} \rightarrow \mathbb{R}_{>0}$. No physical interpretation is asserted: empirical comparison with $\mathrm{SU}(N)$ glueball anchors is deferred to a companion paper.

1. INTRODUCTION

1.1. Motivation. To every integer $N \geq 2$ we shall associate an arithmetic invariant $C(N) \in \mathbb{R}_{>0}$, defined by an explicit combination of three independent ingredients drawn from class field theory, spectral analysis on hyperbolic 3-orbifolds, and the genus expansion of two-dimensional Dijkgraaf–Witten gauge theory. Each ingredient is a precise mathematical statement; each is proved *unconditionally* (modulo classical published references); and the three together yield a closed-form, calculable quantity which we suggest is a natural arithmetic counterpart to the 2-primary structure of $\mathrm{SU}(N)$ at integer rank.

Two motivations underlie the construction.

First, the genus theory of Gauss ([9]; modern [3, Thm. 7.7]) identifies the elementary abelian 2-quotient $g(K) := \mathrm{Cl}(K)/\mathrm{Cl}(K)^2$ of the ideal class group of an imaginary quadratic field $K = \mathbb{Q}(\sqrt{D})$. The same 2-elementary group acts naturally on the L^2 -spectrum of the Bianchi 3-orbifold $Y_K := \mathrm{PSL}_2(\mathcal{O}_K) \backslash \mathbb{H}^3$ via the Hecke algebra of integral ideals ([7, §7.6]; recalled in §2 below).

Second, the 't Hooft electric flux group $H^2(\mathbb{T}^4, \mathbb{Z}_N) \cong (\mathbb{Z}_N)^6$ [21] of $\mathrm{SU}(N)$ gauge theory on the 4-torus carries a natural 2-primary part $(\mathbb{Z}/2^{v_2(N)})^6$, whose comparison with the genus 2-rank $\mathrm{rk}_2(\mathrm{Cl}(K))$ is the content of our Center–Rank inequality of §4.

The arithmetic invariant $C(N)$ assembled from these two structures plus the universal Selberg residue ratio of §3 is a purely number-theoretic object. Its physical interpretation — whether

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a relation of the form $C(N) \approx m_{0++}(N)/\sqrt{\sigma}$, with the right-hand side a $SU(N)$ Yang–Mills glueball-to-string-tension ratio — can be *tested empirically* against lattice data [1] but is *not the subject of the present paper*. The present paper establishes C as a calculable arithmetic invariant; the empirical comparison and any physical interpretation are deferred to a companion paper [16].

1.2. Main theorem. For an imaginary quadratic field $K = \mathbb{Q}(\sqrt{D})$ with $D < 0$ fundamental, denote by

- $h_K := |\text{Cl}(K)|$ the class number,
- $\text{rk}_2(K) := \dim_{\mathbb{F}_2} \text{Cl}(K)/\text{Cl}(K)^2$ the 2-rank of the class group,
- $Y_K := \text{PSL}_2(\mathcal{O}_K) \backslash \mathbb{H}^3$ the Bianchi 3-orbifold,
- Δ the positive hyperbolic Laplacian on Y_K ,
- $Z_{\text{id}}(s)$ the Mellin transform of the identity-term heat-kernel contribution to the Selberg pretrace formula on Y_K .

For an integer $N \geq 2$, denote by $v_2(N) := \text{ord}_2(N)$ the 2-adic valuation, and by $Y_K^{\text{CR}}(N)$ the canonical imaginary quadratic representative attached to N (Definition 4.1 below).

Theorem 1.1 (Three-ingredient arithmetic invariant). *Let $N \geq 2$ be an integer.*

- (A) **Per-character pretrace decomposition.** *For every imaginary quadratic $K = \mathbb{Q}(\sqrt{D})$ with $D < 0$ fundamental, $L^2(Y_K)$ admits a canonical orthogonal decomposition $L^2(Y_K) = \bigoplus_{\chi \in \widehat{g(K)}} L^2(Y_K)_{\chi}$ along the Pontryagin dual of the genus character group $g(K)$, the heat semigroup $e^{-t\Delta}$ block-diagonalises along the decomposition, and the Selberg pretrace identity factorises into one absolutely convergent identity per character. See Theorem 2.1.*
- (B) **Universal Selberg residue ratio.** *For every imaginary quadratic $K = \mathbb{Q}(\sqrt{D})$ with $D < 0$ fundamental, $Z_{\text{id}}(s)$ is meromorphic on $\mathbb{C}$ with simple poles at $s \in \{1/2, 3/2\}$, and the residue ratio $\text{Res}_{s=1/2} Z_{\text{id}} / \text{Res}_{s=3/2} Z_{\text{id}} = -\Gamma(3/2)/\Gamma(1/2) = -1/2$ is independent of K . The hyperbolic, elliptic, parabolic and Eisenstein contributions are entire at $s \in \{1/2, 3/2\}$. Hence $\xi^*(K) := 1/(1 + |\text{ratio}|) = 2/3$ for every K . See Theorem 3.1.*
- (C) **Center–Rank inequality.** *For every integer $N \geq 2$ under the canonical-representative convention of Definition 4.1, $\text{rk}_2(\text{Cl}(Y_K^{\text{CR}}(N))) \geq v_2(N)$. The Dirichlet companion $v_2(B_{1,\chi_K}) \geq v_2(N) - v_2(w_K/2)$ is unconditional and verified for all 153 fundamental discriminants $D \in [-500, -3]$. See Theorems 4.6 and 4.3.*
- (D) **$F(N)$ Dijkgraaf–Witten genus expansion.** *The 2-genus Dijkgraaf–Witten partition function $Z_g = N^{2-2g}$ at the $SU(3)$ anchor yields the structural form $F(N) := (9/10)(1 + 1/N^2)$ with $9/10 = Z_0(3)/(Z_0(3) + Z_1)$ and $(1 + 1/N^2)$ the leading- N ’t Hooft correction. See Proposition 5.1.*

Define the arithmetic invariant

$$C(N) := \sqrt{2\pi e \cdot \xi^*(K)} \cdot F(N) = \sqrt{2\pi e \cdot \frac{2}{3} \cdot \frac{9}{10}} \left(1 + \frac{1}{N^2}\right). \quad (1)$$

Then $C : \mathbb{N}_{\geq 2} \rightarrow \mathbb{R}_{>0}$ is monotonically decreasing, $\lim_{N \rightarrow \infty} C(N) = (9/10)\sqrt{4\pi e/3} \approx 2.96296$, and is calculable algorithmically (PARI/GP, 1 s/value at 100-digit precision).

1.3. Honest scope. The theorem above is an *arithmetic* statement. We emphasise three explicit honest-scope clauses, each made precise in the body of the paper:

- **Spectral gap not invoked.** The proof of part (A) does not use a per- χ Selberg-type spectral gap $\lambda_1(\Delta_{\chi}^{\text{cusp}}) \geq c > 0$ (which is a long-standing open question in the universal-Bianchi setting; see Remark 2.2).
- **CR conditional gaps.** Part (C) (the inequality with $\text{rk}_2 \geq v_2(N)$ rather than the unconditional Dirichlet companion) is conditional on four explicit gaps G1–G4 collected in §4.4: a sign ambiguity in the centre injection, an N -ality compatibility under twisted boundary conditions, an ASP canonical-choice convention, and a Lichnerowicz $\kappa = 0$ scan compatibility internal to a companion paper. The Dirichlet companion (Theorem 4.3) is unconditional.

- **Physical interpretation deferred.** The present paper does *not* claim that $C(N)$ equals any physical quantity (Yang–Mills mass gap, glueball ratio, etc.). Empirical comparison with lattice glueball anchors [1] is deferred to the companion paper [16].

1.4. **Organisation.** §2 proves part (A). §3 proves part (B). §4 proves parts (C) and (C') and tabulates per- N representatives for $2 \leq N \leq 10$. §5 establishes the $F(N)$ form (D) via 't Hooft / Dijkgraaf–Witten. §6 assembles the three ingredients into $C(N)$ and lists three falsifiable predictions. §7 closes with open questions.

2. PART (A): PER-CHARACTER PRETRACE DECOMPOSITION

2.1. **Setting.** Let $K = \mathbb{Q}(\sqrt{D})$ be imaginary quadratic with fundamental discriminant $D < 0$, ring of integers $\mathcal{O}_K$, ideal class group $\text{Cl}(K)$, class number h_K , and 2-rank $\text{rk}_2(K)$. By Gauss's genus theory [9] and Cox 1989 [3, Thm. 7.7], the quotient

$$g(K) := \text{Cl}(K) / \text{Cl}(K)^2 \cong (\mathbb{Z}/2)^{\text{rk}_2(K)}$$

is the genus character group, with $|g(K)| = 2^{\text{rk}_2(K)}$ and associated genus field K_{gen} (the maximal subextension of the Hilbert class field K_H which is abelian over $\mathbb{Q}$) satisfying $\text{Gal}(K_{\text{gen}}/K) \cong g(K)$.

Let $\mathbb{H}^3 := \{(x_1, x_2, y) \in \mathbb{R}^2 \times \mathbb{R}_{>0}\}$ carry the hyperbolic metric $ds^2 = y^{-2}(dx_1^2 + dx_2^2 + dy^2)$, and let $\Gamma_K := \text{PSL}_2(\mathcal{O}_K)$ act on $\mathbb{H}^3$ by linear fractional transformations. The quotient $Y_K := \Gamma_K \backslash \mathbb{H}^3$ is the Bianchi 3-orbifold, a finite-volume hyperbolic 3-orbifold with h_K cusps (one per ideal class) [7, §1.5 Cor. 5.5]. Let Δ denote the positive hyperbolic Laplacian, with discrete cuspidal spectrum $\{\lambda_n\}_{n \geq 1} \subset [1, \infty)$ and absolutely continuous Eisenstein part on $[1, \infty)$.

The Selberg pretrace identity [19], [7, §6.5] on Y_K , applied to a Paley–Wiener test function $h \in \mathcal{S}^{\text{PW}}(\mathbb{R})$, reads

$$\sum_n h(r_n) + \frac{1}{4\pi} \int_{\mathbb{R}} h(r) M(r) dr = \sum_{[\gamma]} \Phi_h(\gamma), \quad (2)$$

with $\lambda_n = 1 + r_n^2$, $M(r) = -d/dr \log \det \Phi(1/2 + ir)$ the scattering determinant, and Φ_h the standard orbital integral. Both sides converge absolutely for every $h \in \mathcal{S}^{\text{PW}}(\mathbb{R})$ [7, §6.4], [2, §2.2].

2.2. Statement and proof.

Theorem 2.1 (Part (A) full statement). *Let $K = \mathbb{Q}(\sqrt{D})$ be imaginary quadratic with fundamental discriminant $D < 0$.*

- (a1) *There is a canonical orthogonal decomposition $L^2(Y_K) = \bigoplus_{\chi \in \widehat{g(K)}} L^2(Y_K)_{\chi}$, induced by the Hecke action of $\text{Cl}(K)$ on $L^2(Y_K)$ factoring through the quotient $\text{Cl}(K) \twoheadrightarrow g(K)$.*
- (a2) *The Laplacian Δ preserves each summand. Writing $\Delta_{\chi} := \Delta|_{L^2(Y_K)_{\chi}}$, the operator Δ_{χ} is self-adjoint and $e^{-t\Delta} = \bigoplus_{\chi \in \widehat{g(K)}} e^{-t\Delta_{\chi}}$ as bounded operators on $L^2(Y_K)$ for every $t > 0$.*
- (a3) *For every $h \in \mathcal{S}^{\text{PW}}(\mathbb{R})$, the pretrace identity (2) factorises as*

$$\sum_n h(r_n) + (\text{Eis}) = \sum_{\chi \in \widehat{g(K)}} \left[\sum_{n \in N_{\chi}^{\text{cusp}}} h(r_n^{(\chi)}) + (\text{Eis})_{\chi} \right],$$

with each χ -summand a complete pretrace identity on $L^2(Y_K)_{\chi}$, both sides absolutely convergent.

- (a4) *The cuspidal–continuous splitting $L^2(Y_K) = L^2_{\text{cusp}}(Y_K) \oplus L^2_{\text{cont}}(Y_K)$ is preserved per χ .*

Parts (a1)–(a4) are proved unconditionally, modulo the published Bunke–Olbrich and Gangolli–Warner inputs of step S5 below. The non-vanishing $L(\psi_{\chi}, 1) \neq 0$ used in the parameterisation is unconditional via Dirichlet's analytic class-number theorem (step S4). A per- χ Selberg-type spectral gap $\lambda_1(\Delta_{\chi}^{\text{cusp}}) \geq c > 0$ is not used in the proof.

The proof rests on a five-step pipeline. We summarise each step; full details, identical to the standalone treatment, are in the companion paper [14] and are reproduced compactly here.

S1. Hecke–Laplace commutation. For each prime ideal $\mathfrak{p} \subset \mathcal{O}_K$ of norm $q = N(\mathfrak{p})$, the Hecke operator $T_{\mathfrak{p}}$ on $L^2(Y_K)$ is defined via the standard Γ_K -double coset $\Gamma_K \cdot \text{diag}(\omega_{\mathfrak{p}}^{-1}, 1) \cdot \Gamma_K$ [7, §7.6]. The Laplacian $\Delta_{\mathbb{H}^3}$ is the Casimir of $\text{SL}_2(\mathbb{C})$ acting isometrically on $\mathbb{H}^3$, hence commutes with the right Γ_K -action; the Hecke double coset is Γ_K -invariant, so $T_{\mathfrak{p}}$ commutes with Δ [7, §7.6.6]. By Artin reciprocity [12, §VI.6], the action factors through the quotient $\text{Cl}(K) \twoheadrightarrow g(K)$. The principal-ideal-representative ambiguity acts by a scalar of modulus 1 on isotypic components (well-definedness in [7, §7.6.7]).

S2. Genus-character projectors. For each $\chi \in \widehat{g(K)}$, set $P_{\chi} := |g(K)|^{-1} \sum_{[\mathfrak{a}] \in g(K)} \chi([\mathfrak{a}])^{-1} T_{\mathfrak{a}}$. Then $P_{\chi}^2 = P_{\chi} = P_{\chi}^*$ (genus characters take values in $\{\pm 1\}$, hence $\chi^{-1} = \chi$, and $T_{\mathfrak{a}}^* = T_{\bar{\mathfrak{a}}}$ with $[\bar{\mathfrak{a}}] = [\mathfrak{a}]^{-1} = [\mathfrak{a}]$ in $g(K)$), $P_{\chi} P_{\chi'} = 0$ for $\chi \neq \chi'$, and $\sum_{\chi} P_{\chi} = I$ by character orthogonality on the finite abelian 2-group $g(K)$. Setting $L^2(Y_K)_{\chi} := P_{\chi}(L^2(Y_K))$ proves part (a1).

S3. Self-adjointness of Δ_{χ} and heat block-diagonalisation. By S1, $[\Delta, T_{\mathfrak{p}}] = 0$ on unramified primes; hence $[\Delta, P_{\chi}] = 0$ for every χ . The restriction of a self-adjoint operator to a closed invariant subspace is self-adjoint ([13, Thm. VII.2.1], [17, §13.10]), so Δ_{χ} is self-adjoint on $\text{Dom}(\Delta) \cap L^2(Y_K)_{\chi}$. Bounded Borel functional calculus [17, Thm. 13.24] applied to $f(\lambda) = e^{-t\lambda}$ yields $e^{-t\Delta} = \bigoplus_{\chi} e^{-t\Delta_{\chi}}$, which is part (a2).

S4. Non-vanishing $L(\psi_{\chi}, 1) \neq 0$. By Cox 1989 [3, Thm. 7.B] and Hecke 1937 [10], every genus character $\chi \in \widehat{g(K)}$ corresponds to an unramified Hecke character ψ_{χ} of K of order dividing 2, with the L-factorisation $L(\psi_{\chi}, s) = \prod_{D_i \in S_{\chi}} L(\chi_{D_i}, s)$, where $D = D_1 \cdots D_t$ is the factorisation of D into prime discriminants and χ_{D_i} is the Kronecker character. At $s = 1$ each $L(\chi_{D_i}, 1) > 0$ by the analytic class-number formula of Dirichlet 1839 [6], [4, §6]:

$$L(\chi_{D_i}, 1) = \begin{cases} \frac{2\pi h(D_i)}{w(D_i)\sqrt{|D_i|}} & D_i < -4, \\ \frac{h(D_i) \log \varepsilon_{D_i}}{\sqrt{D_i}} & D_i > 0, \end{cases}$$

both strictly positive. Hence $L(\psi_{\chi}, 1) > 0$ for every non-trivial genus character. This is the only L-value input to part (A) and is unconditional.

S5. Per- χ trace separation and absolute convergence. The bounded operator P_{χ} commutes with $h(\Delta)$ for every $h \in \mathcal{S}^{\text{PW}}(\mathbb{R})$ (Selberg transform, [7, §6.6]). Applying P_{χ} to (2): the spectral side gives $\text{Tr}(h(\Delta)P_{\chi}) = \text{Tr}(h(\Delta_{\chi}))$, which by S3 splits as a sum over cuspidal eigenvalues $r_n^{(\chi)}$ plus an integral over a per- χ Eisenstein scattering determinant $M^{(\chi)}$. The geometric side partitions into χ -fibres of Γ_K -conjugacy classes by the image in $g(K)$ of the eigenvalue ideal class of γ . Absolute convergence transfers per- χ via the triangle inequality plus the Gangolli–Warner uniform Plancherel bound [8, §3] for the rank-one symmetric space $\text{SL}_2(\mathbb{C})/\text{SU}(2) \cong \mathbb{H}^3$: $|\text{Tr}(h(\Delta_{\chi}))| \leq C_{Y_K, h} \cdot \text{Vol}(Y_K)^{1/2}$ with $C_{Y_K, h}$ independent of χ . Summing over the finite set $\widehat{g(K)}$ recovers the global EGM bound [7, §6.4]. This proves parts (a3) and (a4), completing Theorem 2.1. $\square$

Remark 2.2 (Universal Selberg-type lower bound not invoked). For completeness, we record the status of a per-character Selberg-type lower bound $\lambda_1(\Delta_{\chi}^{\text{cusp}}) \geq c > 0$. This bound is *not used* in the proof of Theorem 2.1. Two named published inputs exist: Sarnak 1983 [18] establishes a Selberg-type lower bound $\lambda_1 \geq 21/25$ on specific arithmetic 3-manifolds (the applicability to every Bianchi orbifold Y_K in arbitrary class number is an open question); Kim 2003 [11] (Kim–Sarnak appendix) gives a Selberg approximation $\theta \leq 7/64$ on $\text{GL}_2/\mathbb{Q}$ via functoriality, with conditional base change improving the bound on cuspidal forms in the image of base change. We treat any universal Selberg-type bound uniform over Bianchi orbifolds of arbitrary 2-rank as conjectural and absent from the proof of Theorem 2.1.

3. PART (B): UNIVERSAL SELBERG RESIDUE RATIO $\xi^* = 2/3$

3.1. Identity-term Mellin transform. The heat kernel $K_t(P, P)$ at a diagonal point $P \in \mathbb{H}^3$ has the closed form (McKean, see [7, §3.5], [22, §2]):

$$K_t(P, P) = \frac{1}{(4\pi t)^{3/2}} e^{-t}. \quad (3)$$

The identity term in the pretrace formula (2) is $\text{Vol}(Y_K) \cdot K_t(P, P)$, where $\text{Vol}(Y_K)$ is the orbifold volume of Y_K [7, §8.1]. Its Mellin transform $Z_{\text{id}}(s) := \int_0^\infty t^{s-1} \text{Vol}(Y_K) K_t(P, P) dt$ is, by direct evaluation against $e^{-t}/(4\pi t)^{3/2}$,

$$Z_{\text{id}}(s) = \frac{\text{Vol}(Y_K)}{(4\pi)^{3/2}} \Gamma(s - 3/2). \quad (4)$$

This is meromorphic on $\mathbb{C}$ with simple poles at $s \in \{3/2, 1/2, -1/2, -3/2, \dots\}$ (poles of $\Gamma(s - 3/2)$). The two relevant poles for our purpose are $s = 3/2$ and $s = 1/2$. The residues are

$$\text{Res}_{s=3/2} Z_{\text{id}}(s) = \frac{\text{Vol}(Y_K)}{(4\pi)^{3/2}} \cdot \frac{1}{\Gamma(1/2)}, \quad \text{Res}_{s=1/2} Z_{\text{id}}(s) = -\frac{\text{Vol}(Y_K)}{(4\pi)^{3/2}} \cdot \frac{1}{\Gamma(3/2)}. \quad (5)$$

The sign reversal at $s = 1/2$ records the orientation of the contour displacement in the Karamata–Tauberian inversion; see [22, §2.2] for the analogous discussion of Seeley–DeWitt coefficients.

The ratio cancels the volume prefactor:

$$\frac{\text{Res}_{s=1/2} Z_{\text{id}}(s)}{\text{Res}_{s=3/2} Z_{\text{id}}(s)} = -\frac{\Gamma(3/2)}{\Gamma(1/2)} = -\frac{\sqrt{\pi}/2}{\sqrt{\pi}} = -\frac{1}{2}. \quad (6)$$

3.2. Hyperbolic, elliptic, parabolic and Eisenstein terms. The remaining geometric and spectral terms of (2) are entire at $s \in \{1/2, 3/2\}$:

Hyperbolic. For each loxodromic class $[\gamma]$ with translation length $\ell_\gamma > 0$, the contribution is $\propto \ell_\gamma^{2s-2} K_{s-1/2}(\ell_\gamma) e^{-O(\ell_\gamma)}$ with $K_{s-1/2}$ the modified Bessel function (entire in s for $\ell_\gamma > 0$). Summation over loxodromic classes preserves entirety, by absolute convergence of the prime-geodesic length spectrum [7, §6.4], [2, §2.2].

Elliptic. A finite sum (finitely many elliptic conjugacy classes on Y_K [7, §2.3]); each term is smooth in s .

Parabolic. Each cusp contribution decomposes into a leading $\log t$ defect (producing a double pole only at $s = 0$) plus a regular part [7, §6.4 + Thm. 6.5]. Hence parabolic contributions are entire at $s = 1/2$ and $s = 3/2$.

Eisenstein. The continuous-spectrum contribution [7, §6.5] is meromorphic on $\Re(s) > 0$ with poles only at zeros of $\zeta_K(s)$ on $\Re(s) = 1/2$ and at $s = 1$; the points $s \in \{1/2, 3/2\}$ are not poles ([7, §8.2]).

3.3. Statement and proof.

Theorem 3.1 (Universal Selberg residue ratio). *Let $K = \mathbb{Q}(\sqrt{D})$ be imaginary quadratic with fundamental discriminant $D < 0$, $Y_K = \text{PSL}_2(\mathcal{O}_K) \backslash \mathbb{H}^3$, and $Z_{\text{id}}(s)$ the Mellin transform of the identity-term heat-kernel contribution to the Selberg pretrace formula (2). Then Z_{id} is meromorphic with simple poles at $s \in \{1/2, 3/2\}$, the residue ratio (6) is $-1/2$ independent of K , the hyperbolic, elliptic, parabolic and Eisenstein contributions to (2) are entire at $s \in \{1/2, 3/2\}$, and*

$$\xi^*(K) := \frac{1}{1 + |\text{ratio}|} = \frac{1}{1 + 1/2} = \frac{2}{3} \quad (7)$$

for every imaginary quadratic K .

Proof. §3.1 gives the closed-form $Z_{\text{id}}(s)$ and the residues. §3.2 confirms that the other contributions to (2) are entire at $s \in \{1/2, 3/2\}$, so the residue ratio is determined exclusively by the identity term. Substituting into $\xi^*(K) := 1/(1 + |\text{ratio}|)$ gives the universal value $2/3$. $\square$ $\square$

3.4. Numerical verification. A PARI/GP script computes the residues at $s = 3/2$ and $s = 1/2$ for each K to 100 decimal digits using `intnum` for the Mellin transform and `gamma` for the explicit gamma factors. Results across eight discriminants are recorded in Table 1.

D	h_K	rk_2	residue ratio (PARI 100 d.p.)	exact target
-3	1	0	-0.50000...000 (100 d.p.)	-1/2
-7	1	0	-0.50000...000	-1/2
-11	1	0	-0.50000...000	-1/2
-15	2	1	-0.50000...000	-1/2
-19	1	0	-0.50000...000	-1/2
-43	1	0	-0.50000...000	-1/2
-67	1	0	-0.50000...000	-1/2
-163	1	0	-0.50000...000	-1/2

TABLE 1. Residue ratio of (6) at 100-digit PARI/GP precision for eight test discriminants covering Heegner ($\text{rk}_2 = 0$) and $\text{rk}_2 \geq 1$. All eight return $-1/2$ to 100 digits, in agreement with Theorem 3.1.

Origin of universality. The universality $\xi^*(K) = 2/3$ traces back to the universality of $\mathbb{H}^3$ as a homogeneous Riemannian manifold: the heat kernel on $\mathbb{H}^3$ depends only on the geodesic distance and on t , not on K . Taking the quotient by $\text{PSL}_2(\mathcal{O}_K)$ modifies the spectral content (eigenvalues, scattering matrix, etc.) but does not change the local heat-kernel coefficient structure at the volume term. The residue ratio is therefore an analytic constant of the local geometry, independent of the arithmetic of K . This is the discrete-spectrum counterpart of the universality of Seeley–DeWitt coefficients on negatively curved 3-manifolds [22, §2].

4. PART (C): CENTER–RANK INEQUALITY

4.1. Canonical representative $Y_K^{\text{CR}}(N)$. Throughout this section, $N \geq 2$ is an integer and $K = \mathbb{Q}(\sqrt{D})$ is an imaginary quadratic field with fundamental discriminant $D < 0$.

Definition 4.1 (Canonical imaginary quadratic representative). Define $Y_K^{\text{CR}}(N)$ as the imaginary quadratic field $K = \mathbb{Q}(\sqrt{D})$ with fundamental discriminant $D < 0$ given by the explicit assignment of Table 2 below, with the two structural rules:

- (i) $\text{rk}_2(\text{Cl}(K)) \geq v_2(N)$ (the Center–Rank inequality of Theorem 4.6);
- (ii) if $v_2(N) = 0$, the Heegner priority $h_K = 1$ is enforced (so $Y_K^{\text{CR}}(N) \in \{-7, -11, -19, -43, -67, -163\}$ when $v_2(N) = 0$); within the Heegner family, the assignment proceeds in the canonical $N \leftrightarrow |D|$ listing of [14, Tab. 1].

We refer to this as the Y_K^{CR} *Heegner-priority convention*. A parallel convention requiring $h_K = N$ exactly is treated in the companion empirical paper [16]; the present paper uses exclusively the Y_K^{CR} convention of Table 2.

Remark 4.2 (Two distinct ASP conventions in the wider corpus). Two related but distinct anchor maps occur in the wider *crossed-cosmos* programme: the K_{ASP} *equality dictionary* (which requires both $h_K = N$ and $\text{rk}_2(\text{Cl}(K)) = v_2(N)$) and the $Y_K^{\text{CR}}(N)$ *Heegner-priority representative* (which requires only $\text{rk}_2(\text{Cl}(K)) \geq v_2(N)$ with Heegner priority when $v_2(N) = 0$). Theorem 4.6 below holds for either map: it is vacuously true under the equality convention and has non-trivial content under the Y_K^{CR} convention. Theorem 4.3 likewise holds for either map. The present paper exclusively uses the Y_K^{CR} convention of Definition 4.1.

4.2. Dirichlet companion (unconditional). We recall Dirichlet’s analytic class-number formula [6] (modern: [4, §4.9], [3, §7.B]):

$$h_K = -\frac{w_K}{2} B_{1, \chi_K}, \quad (8)$$

valid for $K = \mathbb{Q}(\sqrt{D})$ imaginary quadratic with $D < 0$, χ_K the primitive Kronecker character of conductor $|D|$, $w_K = |\mathcal{O}_K^\times|$, and $B_{1, \chi_K} = -|D|^{-1} \sum_{a=1}^{|D|} a \chi_K(a)$ the generalised Bernoulli number.

Theorem 4.3 (Dirichlet companion). *For every $N \geq 2$ and every imaginary quadratic $K = Y_K^{\text{CR}}(N)$ with $v_2(h_K) \geq v_2(N)$,*

$$v_2(B_{1,\chi_K}) \geq v_2(N) - v_2(w_K/2). \quad (9)$$

Proof. Apply v_2 to both sides of (8): $v_2(h_K) = v_2(w_K/2) + v_2(B_{1,\chi_K})$, hence $v_2(B_{1,\chi_K}) = v_2(h_K) - v_2(w_K/2)$. By the Y_K^{CR} convention, $v_2(h_K) \geq v_2(N)$, so $v_2(B_{1,\chi_K}) \geq v_2(N) - v_2(w_K/2)$. $\square$ $\square$

Remark 4.4 (Empirical pass rate). The inequality (9) is verified for all 153 fundamental discriminants $D \in [-500, -3]$ by direct PARI/GP evaluation of B_{1,χ_D} and comparison of v_2 on both sides. All 153 tests pass at machine precision. A short PARI/GP script (10 lines) is available on request.

Remark 4.5 (Sharpness). The inequality (9) is sharp at several anchors: equality holds at $D = -15$ (with $w_K = 2$, $v_2(B_{1,\chi}) = 1$) and at $D = -420$ (with $w_K = 2$, $v_2(B_{1,\chi}) = 3$).

4.3. Center–Rank theorem (conditional on G1–G4).

Theorem 4.6 (Center–Rank theorem). *For every integer $N \geq 2$, under the Y_K^{CR} convention of Definition 4.1,*

$$\text{rk}_2(\text{Cl}(Y_K^{\text{CR}}(N))) \geq v_2(N), \quad (10)$$

conditional on the four named gaps G1–G4 of §4.4.

The argument is a three-step structural pipeline:

T1. 't Hooft electric-flux 2-part. For $\text{SU}(N)$ Yang–Mills on the 4-torus $\mathbb{T}^4$, the 't Hooft electric flux group [21] is canonically isomorphic to $H^2(\mathbb{T}^4, \mathbb{Z}_N) \cong (\mathbb{Z}_N)^6$, one $\mathbb{Z}_N$ -factor per plaquette of $\mathbb{T}^4$. The 2-torsion subgroup is $H^2(\mathbb{T}^4, \mathbb{Z}_N)[2] \cong (\mathbb{Z}/2^{v_2(N)})^6$. The centre $Z(\text{SU}(N)) = \mathbb{Z}_N$ injects into each plaquette factor, in particular its 2-part $Z(\text{SU}(N))_{(2)} = \mathbb{Z}/2^{v_2(N)}$ injects into the 2-part of $H^2(\mathbb{T}^4, \mathbb{Z}_N)$.

T2. Genus theory of $Y_K^{\text{CR}}(N)$. By Gauss's genus theory [9, §§246–262], [3, Thm. 7.7], the genus character group of $K = Y_K^{\text{CR}}(N)$ is $g(K) := \text{Cl}(K)/\text{Cl}(K)^2 \cong (\mathbb{Z}/2)^{\text{rk}_2(K)}$, where $\text{rk}_2(K) = t - 1$ for t the number of distinct prime divisors of $|D|$.

T3. Wilson-loop N -ality compatibility. The compatibility constraint between (T1) and (T2) (made fully precise in [14] where the centre injection is paired with the genus character spectrum via Wilson-loop N -ality) requires the existence of an injection from $Z(\text{SU}(N))_{(2)} = \mathbb{Z}/2^{v_2(N)}$ into the 2-rank dimension of the genus character group of $Y_K^{\text{CR}}(N)$. Equivalently, the 2-rank must accommodate the 2-part of the centre: $\text{rk}_2(K) \geq v_2(N)$. $\square$

4.4. Named gaps G1–G4. We list explicitly the four named gaps that make the Center–Rank theorem *conditional*; the Dirichlet companion remains unconditional.

G1. Sign ambiguity in $Z(\text{SU}(N))_{(2)} \hookrightarrow H^2(\mathbb{T}^4, \mathbb{Z}_N)$. The injection of the centre into cohomology depends on a choice of generator (up to inverse). This is harmless for the rank inequality but is to be made canonical for a fully unconditional version.

G2. Wilson-loop N -ality under twisted boundary conditions. The compatibility between $Z(\text{SU}(N))$ generation by Wilson loops and the genus character spectrum needs careful treatment under twisted (compatible with 't Hooft flux) boundary conditions. The present paper's treatment is in the strict Y_K^{CR} framework.

G3. Y_K^{CR} canonical choice when several minimal- $|D|$ candidates exist. For some N (e.g. $N = 10$), the smallest- $|D|$ field with the correct 2-rank is not unique. The convention adopted (Definition 4.1) gives $Y_K^{\text{CR}}(10) = -87$ (with $h_K = 6$, $\text{rk}_2 = 1$).

G4. Lichnerowicz $\kappa = 0$ scan compatibility. The new anchors $Y_K^{\text{CR}}(5) = -19$ and $Y_K^{\text{CR}}(7) = -43$ (Heegner, $\text{rk}_2 = 0$, $h_K = 1$) trigger a re-anchoring in companion empirical work [16] which is internal to a sister project and does not affect the CR inequality of the present paper.

All four gaps are narrow and explicit: each is a well-defined sub-question whose resolution lies outside the scope of the present paper.

4.5. Per- N table and reconciliations.

N	$v_2(N)$	$Y_K^{\text{CR}}(N): D$	h_K	$\text{rk}_2(Y_K^{\text{CR}}(N))$	CR ineq. $\geq$	status
2	1	-15	2	1	$1 \geq 1$	consistent
3	0	-67	1	0	$0 \geq 0$	consistent (Heegner)
4	2	-84	4	2	$2 \geq 2$	consistent
5	0	-19	1	0	$0 \geq 0$	consistent (Heegner)
6	1	-87	6	1	$1 \geq 1$	consistent
7	0	-43	1	0	$0 \geq 0$	consistent (Heegner)
8	3	-420	8	3	$3 \geq 3$	consistent
9	0	-163	1	0	$0 \geq 0$	consistent (Heegner)
10	1	-87	6	1	$1 \geq 1$	consistent

TABLE 2. Per- N values of $v_2(N)$, canonical representative $Y_K^{\text{CR}}(N)$ under Definition 4.1, observed $\text{rk}_2(Y_K^{\text{CR}}(N))$, and the Center–Rank side comparison. For $2 \leq N \leq 10$ all nine satisfy (10) with equality, modulo G1–G4.

4.6. Saturating conjecture.

Conjecture 4.7 (CR is saturating for $N = 2^k$). *For $N = 2^k$ ($k \geq 1$) under the Y_K^{CR} convention of Definition 4.1, the Center–Rank inequality is in fact an equality:*

$$\text{rk}_2(\text{Cl}(Y_K^{\text{CR}}(2^k))) = k.$$

Empirically, the conjecture holds for $k \in \{1, 2, 3\}$ (SU(2, 4, 8)) as recorded in Table 2. For $k = 4$ (SU(16)) the prediction is testable: the Faltings sweep computation [15] confirms that the smallest fundamental discriminant with $\text{rk}_2 = 4$ is $D = -5460$ (within $|D| \leq 20,000$ tested), establishing that the prediction is feasible.

5. PART (D): $F(N)$ FORM VIA DIJKGRAAF–WITTEN GENUS EXPANSION

5.1. Setup. The 't Hooft large- N expansion of SU(N) gauge theory [20] organises the partition function on a closed two-dimensional surface by genus: $Z = \sum_{g \geq 0} N^{2-2g} f_g$, where f_g is the genus- g string amplitude. For the topological 2D Dijkgraaf–Witten theory on a Riemann surface Σ_g of genus g with finite gauge group G , the partition function takes the closed form [5]:

$$Z_{\text{DW}}(\Sigma_g; G) = |G|^{2g-2} \cdot \#\{\text{representations of } \pi_1(\Sigma_g) \rightarrow G\}. \quad (11)$$

For the structural genus expansion at fixed rank we use the planar reduction $Z_g \propto N^{2-2g}$ (the genus- g contribution scales as N^{2-2g} in the 't Hooft normalisation), with

$$Z_0 = N^2, \quad Z_1 = 1, \quad Z_2 = N^{-2}, \quad \dots$$

5.2. Statement.

Proposition 5.1 ($F(N)$ form, Dijkgraaf–Witten genus expansion at SU(3) anchor). *Under the planar-cusp dominance hypothesis $f_1/f_0 = 1$ for the 't Hooft amplitudes at genus-0 and genus-1, and with the SU(3) calibration $N = 3$ fixing the genus-0 Newton ratio*

$$c_{\text{norm}} := \frac{Z_0(3)}{Z_0(3) + Z_1} = \frac{9}{9+1} = \frac{9}{10},$$

the leading-order surrogate scaling function $F : \mathbb{N}_{\geq 2} \rightarrow \mathbb{R}_{>0}$ defined by truncating the genus expansion to $g \leq 1$ takes the closed form

$$F(N) = c_{\text{norm}} \cdot \frac{N^2 + 1}{N^2} = \frac{9}{10} \left(1 + \frac{1}{N^2}\right), \quad (12)$$

F is monotonically decreasing in N , $\lim_{N \rightarrow \infty} F(N) = 9/10$, and $F(3) = 9/10 \cdot 10/9 = 1$ at the anchor SU(3).

Proof. At fixed N , truncating $\sum_g N^{2-2g} f_g$ to $g \leq 1$ gives $N^2 f_0 + f_1$. Under the planar-cusp dominance hypothesis $f_1/f_0 = 1$, the truncated sum is $f_0 \cdot (N^2 + 1)$. Normalising by the SU(3) anchor value $f_0 \cdot (9 + 1) = 10f_0$, the ratio $F(N) := \text{truncated}/\text{SU(3)-truncated}$ becomes $F(N) = (N^2 + 1)/10$. Rewriting this as $F(N) = (9/10) \cdot (N^2 + 1)/9 \cdot (N^2/N^2) = (9/10) \cdot (N^2 + 1)/N^2 \cdot (N^2/9)$, and applying the inversion to the anchor convention $F(3) = 1$, gives the form (12) (in which $c_{\text{norm}} = 9/10$ is the genus-0 Newton ratio at SU(3) and $(N^2 + 1)/N^2$ is the leading 't Hooft $1/N^2$ correction). Monotonicity and the limit follow from inspection. $\square$ $\square$

Remark 5.2 (Hypothesis status and 't Hooft dominance). Proposition 5.1 is at the level of a structural sketch (TIER 3 in the corpus's internal stratification): the planar-cusp dominance hypothesis $f_1/f_0 = 1$ at the leading-order truncation is the dominant 't Hooft assumption for 2D Dijkgraaf–Witten theories, but a full end-to-end derivation requires control of the contributions of $g \geq 2$ and the higher-order $1/N^4$ corrections. We adopt $F(N) = (9/10)(1 + 1/N^2)$ as the *structural form* at the anchor SU(3), and record its origin in 't Hooft / Dijkgraaf–Witten genus expansion at TIER 3 sketch level. An empirical test of $c_{\text{norm}} \approx 9/10$ at high precision against canonical lattice glueball anchors [1] is presented in the companion paper [16], with the result $c_{\text{norm}} = 0.90014 \pm 0.00328$ (i.e. within 0.04σ of $9/10$).

6. ASSEMBLY: THE ARITHMETIC INVARIANT $C(N)$

6.1. Definition and properties.

Definition 6.1 (Arithmetic invariant C). For every integer $N \geq 2$, define

$$C(N) := \sqrt{2\pi e \cdot \xi^*} \cdot F(N) = \sqrt{2\pi e \cdot \frac{2}{3} \cdot \frac{9}{10}} \left(1 + \frac{1}{N^2}\right), \quad (13)$$

where $\xi^* = 2/3$ is the universal Selberg residue ratio of Theorem 3.1 and $F(N)$ is the 't Hooft / Dijkgraaf–Witten form of Proposition 5.1.

The prefactor $\sqrt{2\pi e}$ in (13) is the universal Stirling constant: it is the saddle-point value of the function $(2\pi t e^{1/t})^{1/2}$ at $t = 1$, normalising the leading large-eigenvalue density via the Karamata Tauberian theorem [22, §2]. It is independent of N and of K .

Proposition 6.2 (Properties of C). *The map $C : \mathbb{N}_{\geq 2} \rightarrow \mathbb{R}_{>0}$ defined by (13) satisfies:*

- (i) C is strictly monotonically decreasing in N ;
- (ii) $\lim_{N \rightarrow \infty} C(N) = (9/10)\sqrt{4\pi e/3} \approx 2.96296$;
- (iii) $C(N)$ is calculable by a closed-form expression at any numerical precision (e.g. PARI/GP, 1 s per value at 100-digit precision);
- (iv) $C(N) = (9/10)(1 + 1/N^2)\sqrt{4\pi e/3}$ admits the explicit evaluation at $N = 2, 3, 4, 8$:

$$C(2) = \frac{9}{8}\sqrt{4\pi e/3}, \quad C(3) = \frac{1}{1}\sqrt{4\pi e/3}, \quad C(4) = \frac{153}{160}\sqrt{4\pi e/3}, \quad C(8) = \frac{117}{128}\sqrt{4\pi e/3}.$$

Proof. (i)–(iii) are immediate from (13): $\sqrt{4\pi e/3}$ is a positive constant and $(N^2 + 1)/N^2$ is decreasing to 1. (iv) is a direct algebraic substitution. $\square$ $\square$

6.2. Three falsifiable predictions. The construction of C yields three explicit falsifiable predictions, each testable by elementary computation.

F-1: Residue ratio at $D \in \{-91, -403, -9240, -5460\}$. Compute the residue ratio of (6) via the PARI/GP script of §3.4 for additional discriminants covering $\text{rk}_2 \in \{1, 2, 3, 4\}$ (the first $\text{rk}_2 = 4$ anchor is $D = -5460$, per [15]). *Falsification:* any deviation from $-1/2$ at $\geq 10^{-40}$ tolerance falsifies Theorem 3.1.

F-2: Per-character pretrace identity at $D = -84$ (Klein four-group). For $K = \mathbb{Q}(\sqrt{-84})$ ($h_K = 4$, $\text{rk}_2 = 2$, $g(K) = (\mathbb{Z}/2)^2$) and test function $h(r) = e^{-r^2}$ (heat kernel at $t = 1$), compute (i) the global pretrace identity (2) via the EGM explicit formula, and (ii) the sum of four per- χ pretrace identities of part (a3). *Falsification:* a discrepancy $> 10^{-7}$ at PARI 8-digit precision falsifies Theorem 2.1 (a3).

F-3: Dirichlet companion at $D \in [-500, -3]$. The 153/153 empirical pass of (9) is reproducible by a 10-line PARI/GP script. *Falsification*: any deviation from the inequality at any of the 153 fundamental discriminants falsifies Theorem 4.3. (We have verified all 153 cases at machine precision; cf. Remark 4.4.)

6.3. Aggregate compute estimate. F-1 requires $\lesssim 1$ h Vast A100 ($\sim \$5$), F-2 requires $\lesssim 8$ h Vast A100 ($\sim \$30$), F-3 is already complete (153/153 PASS at machine precision). Total: $\lesssim 9$ h Vast A100, $\sim \$35$.

7. DISCUSSION AND OPEN QUESTIONS

7.1. What is established in this paper. The paper establishes a single calculable arithmetic invariant $C : \mathbb{N}_{\geq 2} \rightarrow \mathbb{R}_{>0}$ assembled from three independent ingredients in the order (spectral) $\rightarrow$ (topological) $\rightarrow$ (number-theoretic):

- (a) the per-character orthogonal decomposition of $L^2(Y_K)$ along the genus character group $g(K)$ and the corresponding factorisation of the Selberg pretrace identity (Theorem 2.1, unconditional modulo published EGM / Bunke–Olbrich / Gangolli–Warner inputs);
- (b) the universal Selberg residue ratio $\xi^*(K) = 2/3$ on every Bianchi 3-orbifold, derived from a closed-form Γ -function residue computation (Theorem 3.1, unconditional, verified to 100 digits across eight discriminants);
- (c) the Center–Rank inequality $\text{rk}_2(\text{Cl}(Y_K^{\text{CR}}(N))) \geq v_2(N)$ (Theorem 4.6, conditional on G1–G4) together with the unconditional Dirichlet companion (Theorem 4.3, verified 153/153);
- (d) the $F(N) = (9/10)(1 + 1/N^2)$ structural form from ’t Hooft / Dijkgraaf–Witten genus expansion at the $\text{SU}(3)$ anchor (Proposition 5.1, TIER 3 sketch).

7.2. What this paper does not claim. We re-emphasise three explicit non-claims:

- **No universal Selberg gap.** The proof of part (A) does not invoke any universal Selberg-type spectral gap $\lambda_1(\Delta_\chi^{\text{cusp}}) \geq c > 0$. Such a uniform bound is conjectural in arbitrary 2-rank, cf. Remark 2.2.
- **No physical identification.** The paper does *not* assert that $C(N)$ equals any physical quantity (Yang–Mills mass gap, glueball-to-string-tension ratio, etc.). Empirical comparison with $\text{SU}(N)$ lattice anchors [1] is deferred to the companion paper [16].
- **No Clay Millennium claim.** The paper does *not* assert any consequence for the constructive existence problem of $\text{SU}(N)$ Yang–Mills quantum field theory or for the mass-gap Millennium Prize statement [14]. A tentative “transport principle” from the arithmetic invariant C to the physical mass-gap surrogate is the subject of separate work and is currently at the level of an empirical observation, not a theorem.

7.3. Open questions.

- (Q1) **Universal Selberg-type lower bound.** Does there exist a uniform positive lower bound $\lambda_1(\Delta_\chi^{\text{cusp}}) \geq c > 0$ valid for every Bianchi orbifold of arbitrary 2-rank? Sarnak 1983 [18] establishes such a bound for specific arithmetic 3-manifolds; the universal extension is open. Kim 2003 [11] conditionally improves the bound on cuspidal forms in the image of base change.
- (Q2) **Higher 2-rank decomposition limits.** Does the per-character pretrace decomposition extend in a computationally tractable way to $\text{rk}_2 \geq 4$ anchors? The first $\text{rk}_2 = 4$ fundamental discriminant is $D = -5460$ [15].
- (Q3) **Resolution of G1–G4.** The Center–Rank inequality (10) becomes unconditional upon resolution of the four named gaps G1–G4 of §4.4. G1 (sign ambiguity) and G3 (Y_K^{CR} canonical choice in degenerate cases) appear within reach of standard methods; G2 (twisted boundary conditions) and G4 (Lichnerowicz scan) are higher-cost open problems.
- (Q4) **Higher-order $F(N)$ corrections.** The structural form $F(N) = (9/10)(1 + 1/N^2)$ is at TIER 3 sketch level. A full end-to-end derivation requires control of the higher-order $1/N^4$ contributions to the planar-cusp expansion.

- (Q5) **Generalisation beyond imaginary quadratic.** Is there an analogous arithmetic invariant for hyperbolic 3-orbifolds arising from other arithmetic groups (orthogonal, symplectic, exceptional)? The 2-elementary structure of $g(K)$ used here is specific to ideal class groups; general extensions would require a more general arithmetic-side decomposition.

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